Unified Master Compendium: Dimensionless Invariants and Semiclassical Diagnostic Framework

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AI COLLABORATION & AUTHORSHIP TRANSPARENCY STATEMENT

This theoretical framework, its mathematical derivations, and its manuscript preparation were developed through an interactive human–AI research collaboration between **Luís Pimenta Damásio** and **Google's Gemini AI assistant**. Conceptual direction, physical synthesis, metric architecture, and final editorial verification were directed by the human author, with formal mathematical derivations, LaTeX typesetting, and literature verification generated with the aid of Gemini.

Abstract: This Master Compendium unifies a three-part mathematical foundation and five thematic research monographs establishing a covariant, dimensionless invariant architecture across General Relativity, Quantum Field Theory in Curved Spacetime (QFTCS), and Open Quantum Systems. By formulating physical observables in terms of Zero-Angular-Momentum Observer (ZAMO) proper-time exchange rates, Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) decoherence actions bounded by Bekenstein–Hawking Hilbert-space floors, Seeley–DeWitt heat-kernel conformal purities, and an L_∞ Chebyshev semiclassical breakdown regulator, this framework bridges microphysical quantum phase coherence and macroscopic horizon thermodynamics across 120 orders of magnitude.
Keywords: Semiclassical Gravity, Quantum Field Theory in Curved Spacetime, ZAMO Foliation, Open Quantum Systems, Bekenstein–Hawking Holography, Quantum Extremal Surfaces, Superradiance, Trans-Planckian Regularization.

1. Master Table of Contents

Part I: Foundational Custom-Metric Specifications

1. **Covariant Chronological Kinematics and ZAMO Frame Foliation:** ADM 3+1 and Boyer–Lindquist axisymmetric foliation, ZAMO normal congruence n^μ, relational proper-time dilation ρ_τ, clock rate delta $\Delta\rho_\tau$, wave-crest redshift z, and quantum frequency suppression H_Q.
2. **Open-System Decoherence and Holographic Phase Dynamics:** GKSL master equation dynamics, Caldeira–Leggett thermal scattering Γ_{th}, Penrose–Diósi gravitational self-energy dephasing Γ_{grav}, decoherence action Ω, Bekenstein–Hawking inverse-dimension floor Φ_{floor}, Timeline Integrity Φ_τ, surviving coherent cycles N_{eff}, and the Lloyd–Bekenstein gate bound η_{Lloyd}.
3. **Geometric Purity and Multi-Scale Diagnostic Ratios:** Seeley–DeWitt a_4 trace anomaly normalization and Weyl Conformal Purity χ, Buckingham-Π gravity-to-thermal crossover R_{DP}, Penrose–Zel'dovich rotational extraction gain $E_{Penrose}(\theta)$, and the L_∞ Semiclassical Breakdown Index Ξ_{QG}.

Part II: Thematic Research Monographs

1. **Unitary Black Hole Evaporation, Quantum Extremal Surfaces, and the Page Curve:** Ingoing Vaidya collapse, Bogoliubov mode peeling, Quantum Extremal Surface (Island) extremization, Euclidean replica wormhole saddle exchange, and Kerr–Newman evaporation bounded by Φ_{floor} and Ξ_{QG}.

2. **The Quantum-to-Classical Transition: Environmental Decoherence and Gravitationally Induced Collapse:** Feynman–Vernon influence functionals, Itô stochastic localization, and the $R_{\mathrm{DP}} = 1$ mesoscopic crossover between thermal scattering and gravitational self-energy collapse.
3. **Relational Time and Covariant Foliations in Relativistic Quantum Mechanics:** Dirac–Bergmann constraints, the Wheeler–DeWitt frozen formalism, Page–Wootters conditional probability states, Tomonaga–Schwinger multi-fingered time evolution, and ZAMO worldline synchronization.
4. **Trans-Planckian and Curvature Singularity Regulation in Semiclassical Gravity:** Hadamard point-splitting renormalization, Schwinger proper-time heat kernels, Kempf–Mangano–Mann GUP minimal-length cutoffs, Corley–Jacobson dispersive mode bounce, and Ξ_{QG} horizon membrane regulation.
5. **Rotational Energy Extraction and Ultralight Dark Matter Superradiance:** Newman–Penrose tetrad decoupling, the Teukolsky master equation, Sturm–Liouville Wronskian conservation, Zel'dovich–Starobinsky amplification, and Detweiler's gravitational-atom instability scaling (α_{G}, τ_{inst}).

2. Unified Dimensionless Symbol & Metric Glossary

Symbol	Name	Defining Equation	Domain & Physical Interpretation
$\boldsymbol{\rho_\tau}$	ZAMO Dilation Factor	$(\gamma_{\mathrm{pre}}/\sqrt{N^2}_{\mathrm{pre}}(\theta)) \,/\, (\gamma_{\mathrm{obs}}/\sqrt{(1-C_{\mathrm{obs}})})$	$(0, \infty)$; ratio of observer proper time $d\tau_{\mathrm{obs}}$ to target proper time $d\tau_{\mathrm{pre}}$.
$\boldsymbol{\Delta\rho_\tau}$	Clock Rate Delta	$\rho_\tau - 1$	$(-1, \infty)$; net fractional relativistic clock drift ($z = \Delta\rho_\tau$).
$\boldsymbol{H_Q}$	ZAMO Back-Reaction Factor	$\sqrt{N^2}_{\mathrm{pre}}(\theta) \,/\, \gamma_{\mathrm{pre}}$	$(0, 1]$; combined gravitational and kinematic suppression of Compton frequency.
$\boldsymbol{\Omega}$	Decoherence Action	$(\Gamma_{\mathrm{th}} + \Gamma_{\mathrm{grav}}) \cdot t_{\mathrm{pre}}$	$[0, \infty)$; cumulative environmental and gravitational phase-scattering depth.
$\boldsymbol{\Phi_{\mathrm{floor}}}$	Holographic Coherence Floor	$(1 + \pi \cdot 10^{2\zeta})^{-1}$	$(0, (1+\pi)^{-1}]$; inverse Bekenstein–Hawking Hilbert-space dimension ($\zeta = \log_{10}(L/\ell_{\mathrm{P}})$).
$\boldsymbol{\Phi_\tau}$	Timeline Integrity	$\Phi_{\mathrm{floor}} + (1 - \Phi_{\mathrm{floor}})e^{-\Omega}$	$[\Phi_{\mathrm{floor}}, 1]$; surviving quantum state purity bounded by holographic unitarity.
$\boldsymbol{N_{\mathrm{eff}}}$	Surviving Coherent Cycles	$(f_{\mathrm{local}} \cdot t_{\mathrm{pre}}) \cdot \Phi_\tau$	$[0, \infty)$; total usable phase-coherent quantum oscillations prior to thermalization.
$\boldsymbol{\eta_{\mathrm{Lloyd}}}$	Lloyd–Bekenstein Ratio	$N_{\mathrm{eff}} \,/\, (S_{\mathrm{BH}} / k_{\mathrm{B}})$	$[0, \infty)$; coherent cycle utilization relative to holographic bit capacity (≤ 1 bounded).
$\boldsymbol{\chi}$	Weyl Conformal Purity	$(1 + 60A_{\mathrm{W}})^{-1} = (1 + C^2_{\mathrm{clamped}}/\pi^2)^{-1}$	$[\pi^2/(\pi^2+4), 1]$; normalized static vacuum purity under conformal trace anomaly.
$\boldsymbol{R_{\mathrm{DP}}}$	Gravity-to-Thermal Ratio	$\Gamma_{\mathrm{grav}} \,/\, \Gamma_{\mathrm{th}} = (M_{\mathrm{eff}}/M_{\mathrm{P}})^2(\lambda\!\!\!^{-}_{\mathrm{th}}/L)^3$	$[0, \infty)$; ratio of Penrose–Diósi collapse rate to Caldeira–Leggett scattering rate.
$\boldsymbol{E_{\mathrm{Penrose}}(\theta)}$	Penrose–Zel'dovich Gain	$\max(0, 1 - f_{\mathrm{local}}/f_{\mathrm{H}}) \sin^2\theta \cdot \eta_{\mathrm{NT}}(r_{\mathrm{ISCO}})$	$[0, 0.4226]$; latitude-dependent rotational energy extraction efficiency.
$\boldsymbol{\alpha_{\mathrm{G}}}$, $\boldsymbol{\tau_{\mathrm{inst}}}$	Grav. Atom Coupling & Growth	$\alpha_{\mathrm{G}} = M_{\mathrm{eff}} m_0 / M_{\mathrm{P}}{}^2$, $\tau_{\mathrm{inst}} = 24 r_{\mathrm{g}}(1+\chi_{\mathrm{s}})/(c a_* \alpha_{\mathrm{G}}{}^9)$	Dimensionless coupling and Detweiler $2p$ ($\ell{=}m{=}1$) cloud instability timescale.
$\boldsymbol{\Xi_{\mathrm{QG}}}$	Semiclassical Breakdown Index	$\max(\ell_{\mathrm{P}}/L, 10^{-6}/N^2(\theta), (a_*{}^2+Q_*{}^2)/0.9998, T_{\mathrm{eff}}/T_{\mathrm{P}})$	$(0, 1]$; L_∞ Chebyshev regulator monitoring proximity to quantum-gravity breakdown walls.

3. Verified Foundational Bibliography & Acknowledgments

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